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% Problem 1

%implements Gaussian elimination with partial pivoting

```
clc
```

```
A = [ 2  0  1 -1;  
      6  3  2 -1;  
      4  3 -2  3;  
     -2 -6  2 -14];
```

```
b = [6; 15; 3; 12];
```

```
x = Gaussian_elimination_partial_pivoting(A,b);
```

%display the solution vector

```
fprintf('Solution vector\n');  
fprintf('%8.4f\n', x');
```

% Problem 2

%implements Gaussian elimination with scaled partial pivoting

```
clc
```

```
A = [      pi  -exp(1)  sqrt(2) -sqrt(3);  
      pi   exp(1) -exp(1)^2    3/7;  
     sqrt(5) -sqrt(6)         1 -sqrt(2);  
     pi^3 exp(1)^2  -sqrt(7)    1/9];
```

```
b = [sqrt(11); 0; pi; sqrt(2)];
```

```
x = Gaussian_elimination_scaled_partial_pivoting(A,b);
```

%display the solution vector

```
fprintf('Solution vector\n');  
fprintf('%8.4f\n', x');
```

Problem 1

```
function x = Gaussian_elimination_partial_pivoting(A,b)
```

```
    A = [A b];
```

```
    n = size(A,1);
```

```
    for i = 1:n-1
```

```

    %find indices for max value in column of submatrix of interest
    maxindices = find(abs(A(i:n, i)) == max(abs(A(i:n, i))));

    %performs row swap if necessary
    if maxindices(1) ~= 1 % maxindices could be a vector
        temp = A(i, :); % as there may be two maximums
        A(i, :) = A(maxindices(1)+i-1, :);
        A(maxindices(1)+i-1, :) = temp;
        fprintf('Step %i: Swapped row %i and row %i\n', i, i,
maxindices(1)+i-1)
    else
        fprintf('Step %i: No row swap\n', i)
    end

    %performs row ops
    for j = i+1:n
        A(j, i+1:end) = A(j, i+1:end)-A(j, i)/A(i, i)*A(i, i+1:end);
    end

    %sets values of elements below pivot element to zero
    A(i+1:end, i) = zeros(n-i, 1);
end

fprintf('\n')

%creates a vector to store the solution
x = zeros(n, 1);

%solves for x_n
x(n) = A(n, n+1)/A(n, n);

%uses back substitution to solve for x_i
for i = n-1:-1:1
    x(i) = (A(i, n+1)-dot(A(i, i+1:n), x(i+1:n)))/A(i, i);
end
end

Step 1: Swapped row 1 and row 2
Step 2: Swapped row 2 and row 4
Step 3: No row swap

Solution vector
    2.0000
   -0.0000
    1.0000
   -1.0000

```

Problem 2

```

function x = Gaussian_elimination_scaled_partial_pivoting(A,b)
    %creates column vector for scaling values
    scale = max(abs(A'))';

```

```

A = [A b];

n = size(A,1);

for i = 1:n-1
    %find indices for max value of scaled column elements in submatrix of
    interest
    maxindices = find(abs(A(i:n, i)./scale(i:n)) == max(abs(A(i:n, i))./
    scale(i:n)));

    %performs row swap (and scale element swap!) if necessary
    if maxindices(1) ~= 1
        temprow = A(i, :);
        A(i, :) = A(maxindices(1)+i-1, :);
        A(maxindices(1)+i-1, :) = temprow;
        tempscale = scale(i);
        scale(i) = scale(maxindices(1)+i-1);
        scale(maxindices(1)+i-1) = tempscale;
        fprintf('Step %i: Swapped row %i and row %i\n', i, i,
maxindices(1)+i-1)
    else
        fprintf('Step %i: No row swap\n', i)
    end

    %performs row ops
    for j = i+1:n
        A(j, i+1:end) = A(j, i+1:end)-A(j, i)/A(i, i)*A(i, i+1:end);
    end

    %sets values of elements below pivot element to zero
    A(i+1:end, i) = zeros(n-i, 1);
end

fprintf('\n')

%creates a vector to store the solution
x = zeros(n, 1);

%solves for x_n
x(n) = A(n, n+1)/A(n, n);

%uses back substitution to solve for x_i
for i = n-1:-1:1
    x(i) = (A(i, n+1)-dot(A(i, i+1:n), x(i+1:n)))/A(i, i);
end
end

Step 1: No row swap
Step 2: Swapped row 2 and row 4
Step 3: Swapped row 3 and row 4

Solution vector
    0.5551
   -2.3493

```

-0.4903
2.3786

Published with MATLAB® R2024b